

BRIGHAM CITY 28 WNW, UT, USA

WMO: 725770

Lat: 41.6163N

Lon: 112.5437W

Elev: 1509

StdP: 84.46

Time Zone: -7.00 (NAM)

Period: 07-19

WBAN: 04138

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.7	-13.5	-19.4	0.8	-13.1	-17.0	1.0	-11.8	6.7	1.4	5.5	0.5	0.8	N/A	0.508

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.6	34.2	14.7	33.0	14.3	31.7	13.9	16.8	27.5	16.1	27.8	15.5	27.7	2.9	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.5	11.6	19.9	12.0	10.5	19.8	10.6	9.6	19.9	52.9	27.3	50.8	28.0	48.7	27.9	18.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
5.7	4.9	4.2	DB	-19.6	36.2	3.1	0.8	-21.9	36.8	-23.7	37.3	-25.4	37.8	-27.6	38.4	
			WB	-19.7	18.1	3.1	0.7	-22.0	18.6	-23.8	18.9	-25.5	19.3	-27.8	19.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	8.9	-4.7	-1.6	3.7	7.1	12.1	18.5	23.8	22.4	17.1	8.7	2.4	-4.0	(6)
(7)		DBStd	10.51	4.79	4.58	4.20	4.08	4.19	4.52	2.80	2.91	4.17	4.54	4.53	4.67	(7)
(8)		HDD10.0	1858	457	326	199	105	26	2	0	0	5	76	229	434	(8)
(9)		HDD18.3	3858	715	559	453	336	196	52	2	7	68	298	479	692	(9)
(10)		CDD10.0	1438	0	0	4	19	91	258	428	384	217	37	1	0	(10)
(11)		CDD18.3	397	0	0	0	0	3	59	172	132	29	1	0	0	(11)
(12)		CDH23.3	5636	0	0	0	5	81	930	2349	1758	496	17	0	0	(12)
(13)		CDH26.7	2424	0	0	0	0	11	357	1159	775	119	2	0	0	(13)
(14)	Wind	WSAvg	1.8	1.3	1.7	1.9	2.2	2.1	2.1	2.0	1.8	1.6	1.5	1.3	(14)	
(15)	Precipitation	PrecAvg	354	33	27	30	39	48	29	15	18	24	32	26	33	(15)
(16)		PrecMax	565	86	72	81	90	124	92	49	54	83	125	66	85	(16)
(17)		PrecMin	229	3	0	7	4	2	1	0	1	1	0	6	1	(17)
(18)		PrecStd	91	22	18	19	20	35	23	13	13	18	25	15	21	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.4	12.5	18.9	23.6	28.4	35.0	35.9	34.7	31.8	25.6	16.4	10.4	(19)
(20)			MCWB	2.7	5.3	8.1	10.5	11.6	15.4	14.8	14.6	14.3	12.4	8.5	5.5	(20)
(21)		2%	DB	5.3	9.6	16.6	20.7	25.6	32.6	34.7	33.4	29.6	21.7	14.2	6.9	(21)
(22)			MCWB	2.0	4.1	7.3	8.7	11.5	13.5	14.7	14.3	13.1	10.9	6.9	3.2	(22)
(23)		5%	DB	3.3	7.3	14.1	18.4	23.6	30.5	33.7	32.2	27.9	19.2	12.2	4.2	(23)
(24)			MCWB	0.7	3.0	6.0	8.1	10.7	12.8	14.6	13.9	12.4	9.7	5.9	1.7	(24)
(25)		10%	DB	1.7	5.3	11.6	15.9	21.1	28.7	32.5	31.0	26.4	17.0	9.8	2.4	(25)
(26)			MCWB	-0.3	2.1	4.9	7.2	10.1	12.5	14.4	13.7	12.2	8.8	4.7	0.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.7	6.2	8.9	11.5	13.7	16.3	17.8	17.3	17.1	13.1	9.1	5.9	(27)
(28)			MCDWB	6.2	11.7	17.2	21.0	23.0	32.9	27.6	26.2	23.7	21.4	14.9	9.7	(28)
(29)		2%	WB	2.5	4.9	7.8	10.0	12.2	15.2	16.9	16.5	15.5	12.0	7.3	4.2	(29)
(30)			MCDWB	4.9	9.5	15.8	18.9	22.4	29.1	27.7	27.2	24.9	19.4	13.2	7.1	(30)
(31)		5%	WB	1.2	3.8	6.8	8.8	11.4	14.2	16.4	15.7	14.4	11.0	6.0	2.3	(31)
(32)			MCDWB	3.3	7.4	14.0	16.8	21.4	27.5	27.9	27.3	24.1	17.7	10.9	4.3	(32)
(33)		10%	WB	0.1	2.7	5.6	7.8	10.7	13.4	15.8	15.1	13.4	9.8	4.8	0.9	(33)
(34)			MCDWB	1.7	5.8	11.6	15.1	20.1	26.4	28.1	27.2	23.4	15.7	9.2	2.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.5	8.6	10.6	12.1	12.6	15.2	15.6	14.7	12.6	10.8	8.2	(35)	
(36)			MCDBR	9.1	9.9	13.9	16.3	15.8	17.2	17.0	16.7	14.8	13.0	9.4	(36)	
(37)			MCWBR	6.4	6.1	7.2	7.7	6.6	6.3	6.4	6.5	7.0	7.3	7.3	6.2	(37)
(38)		5% WB	MCDWB	7.9	9.2	13.4	14.1	14.1	16.2	14.8	14.7	13.8	12.8	12.2	8.5	(38)
(39)			MCWBR	5.9	5.8	7.1	7.2	6.5	6.4	5.9	5.6	5.9	6.4	7.3	6.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.241	0.253	0.265	0.294	0.308	0.306	0.330	0.325	0.299	0.271	0.251	0.240	(40)	
(41)		taud	2.530	2.590	2.602	2.500	2.495	2.534	2.472	2.475	2.542	2.616	2.623	2.563	(41)	
(42)		Ebn at Noon	942	980	998	979	966	964	936	934	944	944	926	917	(42)	
(43)		Edh at Noon	69	77	86	102	106	102	108	104	90	74	63	61	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.76	2.67	4.17	5.54	6.57	7.69	7.51	6.57	5.23	3.59	2.31	1.49	(44)	
(45)		RadStd	0.25	0.34	0.30	0.33	0.50	0.48	0.29	0.29	0.32	0.21	0.19	0.22	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page